

EG2131D Chip Datasheet

MOS transistor gate driver chip

Version Change Log

Version number	date, description	
V1.0	EG2131D Datasheet Draft (January 18, 2024)	
V1.1	On March 11, 2026, the VB withstand voltage was changed from 220V to 250V.	

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EG2131D Chip Datasheet V1.0

1. Characteristics

• High-end floating bootstrap power supply design, withstand voltage up to 250V

• With internal bootstrap diode

• Adaptable to 5V and 3.3V input voltages

Supports a maximum frequency of 500kHz .

Low -end VCC undervoltage shutdown output

Output current capability IO+/- 1A/1.5A

Built- in dead-time control circuit

• It has a built-in interlocking function, which completely prevents the simultaneous conduction of the upper and lower tube outputs.

• The HIN input channel is active high, controlling the high-side HO output.

The LIN input channel is active low, controlling the low-side LO output.

Few peripheral components

Package type: SOP8

Lead-free and halogen-free , RoHS compliant

2. Description

The EG2131D is a high-performance, cost-effective dedicated chip for driving the gates of high-power MOSFETs and IGBTs, and it integrates logic signal inputs.

Processing circuit, dead-time control circuit, undervoltage shutdown circuit, latch-up circuit, level shifting circuit, pulse filter circuit, and output drive circuit.

Drive circuit specifically designed for use in brushless motor controllers.

The EG2131D has a high-end operating voltage of up to 250V and a low-end VCC power supply voltage range of 11V to 20V, with a static power consumption of approximately 100uA.

This chip features a latch-up function to prevent simultaneous conduction of the output power transistors. The input channel HIN has a built-in 250K pull-down resistor, and LIN has a built-in upper...

Pulling up the potential turns the upper and lower power MOSFETs off when the input is floating. The output current capability is IO+/- 1/1.5A, and it is packaged in an SOP8 package.

3. Application Areas

Power tools ; portable

power banks; high-voltage fast-charging switching power

supplies ; variable frequency water pump controllers

Electric vehicle control

• Brushless motor driver

4. Pins

4.1 Pin Definitions

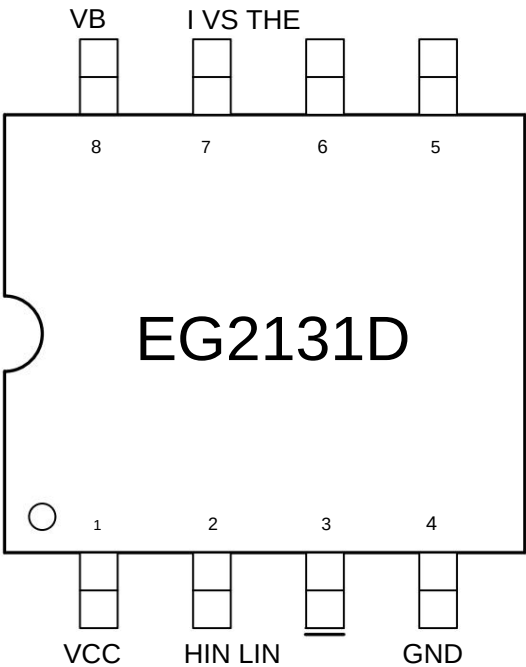


Figure 4-1. EG2131D Pin Definitions

4.2 Pin Description

Pin number	Pin name	I/O	describe
1	VCC	Power	The chip's operating power input has a voltage range of 11V-20V, and an external high-frequency bypass capacitor can... Reduce high-frequency noise at the chip input.
2	DON'T		The logic input control signal is active high, controlling the on/off state of the high-side power MOSFET. end "0" means the power MOSFET is turned off. "1" indicates that the power MOSFET is enabled.
3	LIN		The logic input control signal is active low, controlling the on/off state of the low-side power MOSFET. end "1" means turn off the power MOSFET. "0" indicates that the power MOSFET is enabled.
4	GND	GND is the ground terminal of the chip.	
5	IT	O output controls the on/off state of the low-side MOSFET power transistor.	
6	VS	O High-end suspended ground end	
7	TO	O output controls the on/off state of the high-side MOSFET power transistor.	
8	VB	Power High-end Floating Power Supply	

5. Structural Diagram

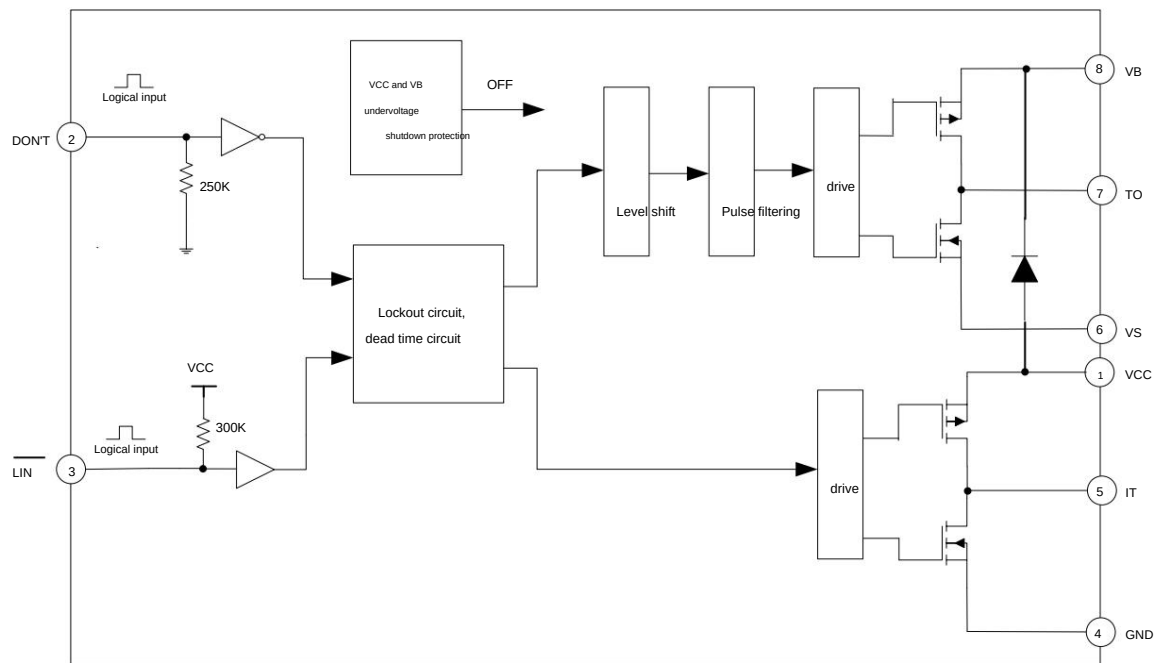


Figure 5-1. Internal circuit diagram of EG2131D

6. Typical Application Circuits

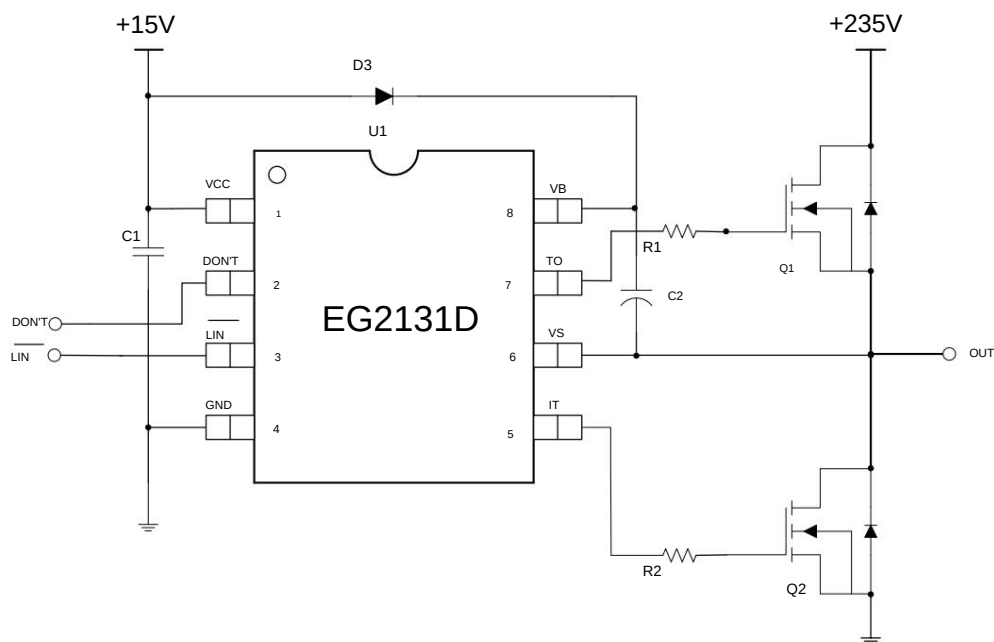


Figure 6-1. Typical application circuit diagram of EG2131D

7. Electrical characteristics

7.1 Limiting Parameters

Unless otherwise specified, under the condition of TA=25℃

Symbol	Parameter Name	Bootstrap	High-end	VB	Test conditions	minimum	and maximum	units	
VB		Power Supply				-0.3		250	In
VS		High-end	suspended	ground		VB-25	VB+0.3 V		
TO		High-end	output			VS-0.3	VB+0.3 V		
IT		Low-end	output			-0.3	VCC+0.3 V		
VCC		High-				-0.3		25	In
DON'T		channel	power	logic	signal	output		-0.3	VCC+0.3 V
		Input	level						
LIN		low	channel	logic	signal	input		-0.3	VCC+0.3 V
		Input	level						
FACING		ambient	temperature			-40		125	℃
Tstr		Storage	temperature			-55		150	℃
TL		welding	temperature		T=10S			300	℃

Note: Exceeding the listed limits may cause permanent damage to the chip's internal components. Prolonged operation under extreme conditions may affect the chip's reliability.

7.2 Typical Parameters

Unless otherwise specified, the conditions are: TA=25℃, VCC=15V, and load capacitance CL=10nF.

Parameter Name	symbol	Test conditions	minimum	typical maximum	unit	
Power Supply	Vcc			11	15	20 In
VCC quiescent current Icc		Input is left blank. VCC=15V			80	a
VB static current	IB input left floating, VB=15V -				30	a
input logic signal high Low	Vin(H) All input control signals 2.5					In
potential input logic signal High	Vin(L) -0.3 for all input control signals.			0	1.5	In
potential input logic signal Low level current	Yes(H)	Vin=5V			20	a
input logic signal Level current	Yes(L)	Vin=0V		-20		a
VCC power supply undervoltage shutdown characteristic						
Vcc (on) is the turn-on voltage.				8.4	9.4	10.4 In
Vcc (off) is the turn-off voltage.				7.8	8.8	9.8 In
VB power supply undervoltage shutdown feature						
VB (on) is the on-state voltage.				7.7	8.7	9.7 In
VB (off) voltage				7.2	8.2	9.2 In
Low-side output LO switching time characteristics						
Enable delay	Ton	See Figure 7-1			350	nS
Guan Yinshi	Toff	See Figure 7-1			150	nS
Ascent Time	Tr	See Figure 7-1			180	nS
Descent time	Tf	See Figure 7-1			70	nS
High-end output HO switching time characteristics						
Enable delay	Ton	See Figure 7-2			350	nS
Guan Yinshi	Toff	See Figure 7-2			150	nS
Ascent Time	Tr	See Figure 7-2			180	nS
Descent time	Tf	See Figure 7-2			70	nS
Dead time characteristics						
Dead Time	DT	See Figure 7-3. No-load capacitor CL=0		100 200 300		nS
Maximum drive capability of IO output						

IO output current IO+		Vo=0V,Vin=VIH PWȳ10uS		1		A
IO output sink current IO-		Vo=12V,VIN=VIL PWȳ10uS		1.5		A

7.3 Switching time characteristics and dead time waveform diagram

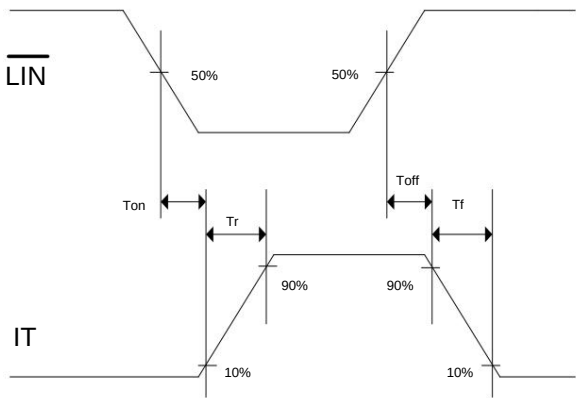
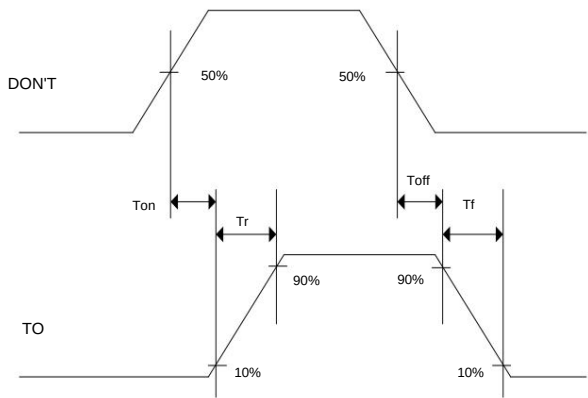


Figure 7-1. Waveform diagram of low-end output (LO) switching time.



7-2. High-side output HO switching time waveform diagram

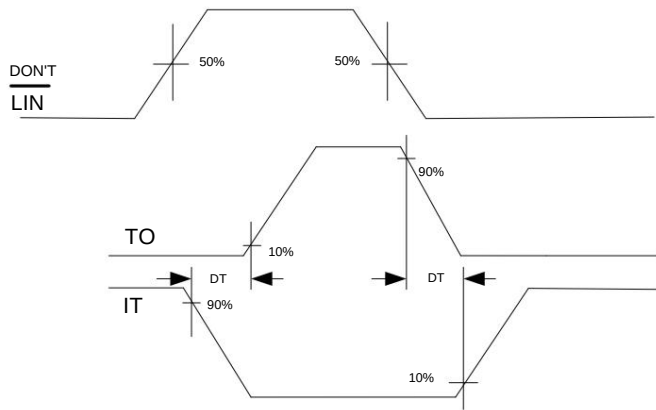


Figure 7-3. Dead time waveform diagram

8. Application Design

8.1 VCC terminal power supply voltage

To ensure sufficient drive voltage for driving N-channel power MOSFETs, a typical operating voltage of 11V-20V for the power supply VCC is recommended.

The ground of the EG2131D chip is the same as the ground of the MCU.

8.2 Input Logic Signal Requirements and Output Driver Characteristics

The EG2131D's main functions include logic signal input processing, dead time control, level conversion, floating bootstrap power supply structure, and upper and lower logic control.

Bridge totem pole output. The logic signal input has a high-level threshold of 2.5V and a low-level threshold of 1.5V. The logic signal input must be...

With its low output current, the MCU's output logic signals can be directly connected to the input channels of the EG2131D.

The high-end upper bridge arm and low-end lower bridge arm output drivers have a maximum sink current of up to 1.5A and a maximum output current of up to 1A. The high-end upper bridge arm...

The circuit can withstand a voltage of 250V, with a small conduction delay between the input logic signal and the output control signal. The low-side output turn-on conduction delay is [missing information].

350ns turn-on conduction delay, 150ns turn-off conduction delay; high-side output turn-on conduction delay is 350ns, low-side output turn-on conduction delay is 150ns.

The rise time is 180ns and the fall time is 70ns. The rise time for high-side output activation is 180ns and the fall time for deactivation is...

70ns

The logic function diagram for the input and output signals is shown in Figure 8-2:

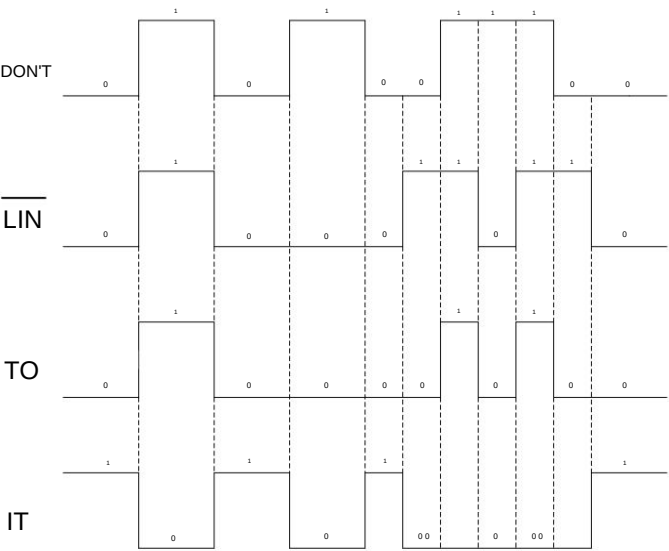


Figure 8-2. Logic function diagram of input and output signals

Input and output signal logic truth tables:

enter		Output	
Input and output logic			
DON'T	LINyyyyy	TO	IT
0	0	0	1
0	1	0	0
1	0	0	0
1	1	1	0

From the truth table, it can be seen that when the input logic signals HIN and LIN are not both "0" and not both "1", the driver controls the output HO.

When LO is simultaneously "0", both the upper and lower power transistors are turned off; when the input logic signals HIN and LIN are simultaneously "0", the driver control output HO is "0".

When the upper transistor is off, LO is "1" and the lower transistor is on; when the input logic signals HIN and LIN are both "1", the driver control output HO is "1" and the upper transistor is on.

When LO is "0", the lower transistor is turned off; the internal logic processor prevents the upper and lower power transistors at the controller output from being turned on simultaneously, and has a mutual interlocking function.

8.3 Bootstrap Circuit

The EG2131D employs a bootstrap floating drive power supply structure, greatly simplifying drive power supply design, requiring only one power supply voltage, VCC.

The driving of two power switching devices, a high-end N-channel MOSFET and a low-end N-channel MOSFET, greatly facilitates practical applications.

The EG2131D can automatically perform bootstrap voltage boosting using an external bootstrap diode (as shown in Figure 8-3) and a bootstrap capacitor. Assuming the lower diode is on...

During the off-state of the upper transistor, the bootstrap capacitor C has been charged to a sufficient voltage ($V_c=V_{CC}$). When HO outputs a high level, the upper transistor is turned on and the lower transistor is turned off, VC

The voltage on the bootstrap capacitor will act as an equivalent voltage source to power the internal drivers VB and VS, thus driving the high-side N-channel MOSFET.

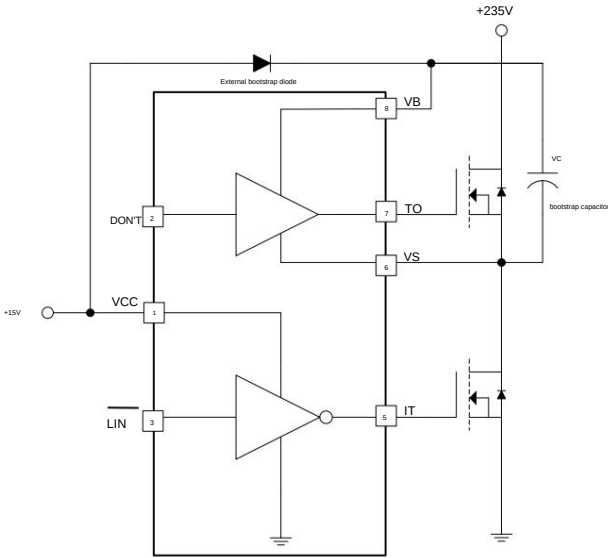


Figure 8-3. EG2131D bootstrap circuit structure

9. Package size

9.1 SOP8 package dimensions

